

Intrinsic Peripheral Residue Transfer from Weak Finite Factors

Direct Haar-Range Bounds and a Sharp Half-Power Barrier

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Abstract

The two-pole Hardy reduction for the noisy quadratic transfer operator left a finite-factor premise: the intrinsic Perron and parity residues must not regain a power-law loss on the adjacent Haar coupling ranges. The previous proposed route asked the finite left eigenfactors to inherit the sharp rounded-spike detail estimate $\|\ell_d\|_2 = O(h\sigma^{-1})$, which would make the fine residues $O(\sqrt{\sigma})$. We prove that this sharp transfer is unnecessary.

The strong/weak Ulam stability inherited from the postcritical tower/spike space gives finite peripheral normalizations satisfying

$$|\lambda_{n,-}| \geq \frac{1}{2}, \quad \|r_{n,\pm}\|_\infty + \|\ell_{n,\pm}\|_1 \leq C$$

on every $h = o(\sigma^2)$ schedule. Direct Gaussian smoothing gives

$$\|(E_{2n} - E_n)\mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} \leq Ch\sigma^{-3/2},$$

$$\|(E_{2n} - E_n)\mathcal{K}_\sigma\|_{L^\infty \rightarrow L^2} \leq Ch\sigma^{-1}, \quad \|\mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} \leq C\sigma^{-1/2}.$$

We also prove $\|B\|_{\mathfrak{S}_2}, \|C\|_{\mathfrak{S}_2} = \Theta(h\sigma^{-3/2})$. The exact block eigenvector identities then yield

$$\frac{\|U^*P_{f,\pm}UB\|_{\mathfrak{S}_2}}{\|B\|_{\mathfrak{S}_2}} = O(1), \quad \frac{\|CP_{c,-}\|_{\mathfrak{S}_2}}{\|C\|_{\mathfrak{S}_2}} = O(1),$$

while $CP_{c,+} = 0$ exactly. Hence the sufficient residue premise in RH-50 closes without a sharp finite-detail theorem and without a polylogarithmic bound for the complete finite parity projector.

There is a precise reason not to demand more from weak factor information: when h/σ is small,

$$\|(E_{2n} - E_n)\mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} = \Theta(h\sigma^{-3/2}).$$

Thus uniform left L^1 control alone cannot produce $O(h\sigma^{-1})$; the missing half power requires the actual postcritical spike geometry.

A five-scale binary64 audit with $N\sigma = 20.48$ reaches $N = 40960$. The parity weak condition product decreases from 1.0968 to 1.0385, and the observed parity detail divided by $h\sigma^{-1}$ stays between 0.0465 and 0.0553. Fine Perron, fine parity, and right parity residues have fitted vanishing powers 0.5275, 0.5500, and 0.9362. These stronger laws are floating evidence, not analytic claims. The remaining Stage A gates are the growing-horizon Hardy-energy budget and infinite-tail/cutoff validation. No arithmetic trace formula, zeta-zero identity, self-adjoint realization, or Riemann-hypothesis conclusion is claimed.

Keywords: Ulam approximation; peripheral eigenfactor; Riesz residue; Haar detail; Gaussian smoothing; strong-weak stability; small-noise transfer operator.

MSC 2020: 37M25; 47A55; 47B65; 65R20; 37A30.

1 Introduction

RH-49 and RH-50 reduced the remaining intrinsic-identification problem to two-pole Hilbert–Schmidt Hardy energies and explicit peripheral residues [5, 6]. Once the Perron and negative-parity poles are removed, the bulk range actions can be studied by Stein/Gramian certificates. RH-51 then showed that exact fixed-rank Stein factors are obstructed, but growing-horizon block certificates remain viable [8].

The present paper addresses the other premise in RH-50. Let $V_{2n} = V_n \oplus W_n$ be the adjacent Haar split and

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

the fine finite Markov matrix. If $P_{f,s} = r_{f,s} \otimes \ell_{f,s}$, the normalized fine residue is

$$\frac{\|U^* P_{f,s} U B\|_{\mathfrak{S}_2}}{\|B\|_{\mathfrak{S}_2}}.$$

The original route sought $\|\ell_{f,s,d}\|_2 = O(h\sigma^{-1})$, because $\|B\|_{\mathfrak{S}_2} = \Theta(h\sigma^{-3/2})$ then gives $O(\sqrt{\sigma})$.

That target is true in the stored data, but it is stronger than the program needs. We instead use the weak norms naturally controlled by Keller–Liverani stability: right L^∞ and left L^1 . The resulting factor-detail bound is only $O(h\sigma^{-3/2})$, yet its power exactly cancels the Hilbert–Schmidt coupling denominator. The coarse right parity residue has a symmetric direct estimate. Both are uniform $O(1)$, already better than the polylogarithmic RH-50 premise.

The distinction is structural. We prove that the $L^1 \rightarrow L^2$ target-detail norm really has scale $h\sigma^{-3/2}$. No proof based only on a uniform L^1 bound can recover the rounded-spike half power. The stronger result must use where the physical eigenfactor places its mass, not merely its total variation.

2 Finite factors and adjacent Haar channels

Let $I = [0, 1]$, $f(x) = 1 - u_c x^2$, and

$$k_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(-y - f(x))}{Z_\sigma(x)}.$$

The observable and density operators are

$$(\mathcal{K}_\sigma v)(x) = \int_I k_\sigma(x, y) v(y) dy, \quad (\mathcal{P}_\sigma q)(y) = \int_I k_\sigma(x, y) q(x) dx.$$

For equal-cell averaging E_n , let

$$G_{n,\sigma} = E_n \mathcal{K}_\sigma E_n|_{V_n}.$$

Its density matrix adjoint is the restriction to V_n of

$$\mathcal{P}_{n,\sigma} = E_n \mathcal{P}_\sigma. \tag{1}$$

The source projection is unnecessary because each finite eigendensity already belongs to V_n .

For $V_{2n} = V_n \oplus W_n$, write

$$G_{2n,\sigma} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

The coarse cell width is $h = 1/n$. For $s \in \{+, -\}$,

$$G_{m,\sigma} r_{m,s} = \lambda_{m,s} r_{m,s}, \quad G_{m,\sigma}^* \ell_{m,s} = \overline{\lambda_{m,s}} \ell_{m,s}, \quad \langle \ell_{m,s}, r_{m,s} \rangle = 1.$$

Right vectors are cell values and left vectors are cell masses. Thus

$$\|\ell\|_1 = \sum_j |\ell_j|, \quad \|\ell\|_2 = \sqrt{m} \|\ell\|_{\ell^2}, \quad \|r\|_\infty = \max_j |r_j|, \quad \|r\|_2 = m^{-1/2} \|r\|_{\ell^2}.$$

3 Uniform weak finite factors

The continuum spectral stability was established on the postcritical tower/spike space $\mathcal{B}$, with L^1 as weak norm [4, 7]. The deterministic peripheral eigenvalues 1 and -1 are simple and uniformly isolated from the remaining strong-space spectrum.

Lemma 3.1 (Equal-cell approximation on the spike space). *There is $C < \infty$ such that*

$$\|E_n q - q\|_{L^1} \leq C\sqrt{h} \|q\|_{\mathcal{B}}. \quad (2)$$

On the regular bounded-variation component this improves to $Ch \|q\|_{\text{BV}}$. Conditional expectation is L^1 -contractive and variation diminishing after a density transfer step.

Proof. The regular estimate is the standard cellwise bounded-variation bound [1, 3]. For a normalized one-sided square-root atom $a(t-s)_+^{-1/2} \chi(t-s)$, the cell containing s has L^1 mass $O(\sqrt{h})$. Away from it, summing the derivative bound gives

$$h \int_h^\delta t^{-3/2} dt = O(\sqrt{h}).$$

There are finitely many postcritical atoms, with coefficients controlled by the $\mathcal{B}$ -norm. Conditional expectation is an L^1 contraction and does not increase variation of the regular representative. $\square$

Theorem 3.2 (Uniform weak Ulam peripheral factors). *Let $\sigma \downarrow 0$ and $h = h(\sigma)$ satisfy*

$$h = o(\sigma^2). \quad (3)$$

For sufficiently small σ , the finite operators have exactly two simple peripheral branches near 1 and -1 . They can be normalized so that

$$r_{n,+} = \mathbf{1}, \quad \|\ell_{n,+}\|_1 = 1, \quad (4)$$

$$|\lambda_{n,-}| \geq \frac{1}{2}, \quad \|r_{n,-}\|_\infty + \|\ell_{n,-}\|_1 \leq C. \quad (5)$$

Proof. RH-14 gives uniform Lasota–Yorke estimates on $\mathcal{B}$, compact strong-to-weak embedding, and

$$\|\mathcal{P}_\sigma - \mathcal{P}_0\|_{\mathcal{B} \rightarrow L^1} = o(1).$$

By lemma 3.1 and (1),

$$\|\mathcal{P}_{n,\sigma} - \mathcal{P}_\sigma\|_{\mathcal{B} \rightarrow L^1} \leq C\sqrt{h}.$$

The variation-diminishing property preserves the same Lasota–Yorke ledger for $E_n \mathcal{P}_\sigma$. Therefore $\mathcal{P}_{n,\sigma} \rightarrow \mathcal{P}_0$ strongly-to-weakly along (3). Keller–Liverani stability preserves the ranks of the two isolated projectors and gives weak convergence of their ranges [2]. The dual statement applied to the bounded component-sign observable controls the right eigenfunctional. The deterministic left component densities have finite L^1 norm and pair nontrivially with that sign. Biorthogonal normalization gives (5). Perron normalization follows exactly from stochasticity and positivity. $\square$

Remark 3.3. The schedule is precisely $n\sigma^2 \rightarrow \infty$, already adopted for the intrinsic small-noise limit. No L^2 contour-resolvent upper enters the proof. Weak factors stay bounded while their L^2 projector norms may grow logarithmically.

4 Direct smoothing and coupling scales

Put $Q_n = E_{2n} - E_n$.

Lemma 4.1 (Three direct smoothing bounds). *For h/σ sufficiently small,*

$$\|Q_n \mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} \leq Ch\sigma^{-3/2}, \quad (6)$$

$$\|Q_n \mathcal{K}_\sigma\|_{L^\infty \rightarrow L^2} \leq Ch\sigma^{-1}, \quad (7)$$

$$\|\mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} \leq C\sigma^{-1/2}. \quad (8)$$

Proof. For each x , target-cell Poincaré gives

$$\|Q_n k_\sigma(x, \cdot)\|_2 \leq \frac{h}{\pi} \|\partial_y k_\sigma(x, \cdot)\|_2 \leq Ch\sigma^{-3/2}.$$

Minkowski proves (6). For $\|v\|_\infty \leq 1$,

$$|\partial_x \mathcal{K}_\sigma v(x)| \leq \int_I |\partial_x k_\sigma(x, y)| dy \leq C\sigma^{-1}.$$

Source-cell Poincaré proves (7). Finally,

$$\|\mathcal{P}_\sigma q\|_2 \leq \int_I |q(x)| \|k_\sigma(x, \cdot)\|_2 dx \leq C\sigma^{-1/2} \|q\|_1.$$

□

Proposition 4.2 (Two-sided adjacent coupling scales). *There are $c, C, a_0, \sigma_0 > 0$ such that*

$$ch\sigma^{-3/2} \leq \|B\|_{\mathfrak{S}_2}, \|C\|_{\mathfrak{S}_2} \leq Ch\sigma^{-3/2} \quad (9)$$

when $0 < \sigma < \sigma_0$ and $h/\sigma < a_0$.

Proof. The upper bounds are the target/source Poincaré estimates. RH-50 proved the B lower bound. For C , choose an interior source rectangle on which $|f'|$ is bounded below and the row normalizer is exponentially close to one. There

$$\iint |\partial_x k_\sigma(x, y)|^2 dx dy \geq c\sigma^{-3}.$$

The leading source-Haar coefficient is a fixed nonzero first moment times $h\partial_x k_\sigma$. Summing its square gives $ch^2\sigma^{-3}$; the remainder is relative $O(h/\sigma)$. □

5 Direct closure of the peripheral residues

For a fine branch write

$$r_f = \begin{pmatrix} r_c \\ r_d \end{pmatrix}, \quad \ell_f = \begin{pmatrix} \ell_c \\ \ell_d \end{pmatrix}.$$

The left block eigenvector equation gives

$$B^* \ell_c = (\bar{\lambda} - D^*) \ell_d, \quad U^* P_f U B = r_c \otimes ((\bar{\lambda} - D^*) \ell_d). \quad (10)$$

Theorem 5.1 (Uniform direct residue closure). *Along every schedule (3),*

$$\boxed{\frac{\|U^* P_{f,s} U B\|_{\mathfrak{S}_2}}{\|B\|_{\mathfrak{S}_2}} = O(1), \quad s \in \{+, -\}.} \quad (11)$$

On the coarse right,

$$\boxed{CP_{c,+} = 0, \quad \frac{\|CP_{c,-}\|_{\mathfrak{S}_2}}{\|C\|_{\mathfrak{S}_2}} = O(1).} \quad (12)$$

Proof. The finite density eigenrelation and lemma 4.1 give

$$\|\ell_d\|_2 = |\lambda|^{-1} \|Q_n \mathcal{P}_\sigma \ell_f\|_2 \leq Ch\sigma^{-3/2} \|\ell_f\|_1.$$

The last factor, $\|r_c\|_2$, $|\lambda|$, and $\|D\|$ are uniformly bounded by theorem 3.2. Insert this estimate into (10) and divide by the B lower bound.

The Perron right vector is constant and $C\mathbf{1} = 0$. For parity,

$$\|Cr_{c,-}\|_2 \leq Ch\sigma^{-1} \|r_{c,-}\|_\infty$$

by (7), while

$$\|\ell_{c,-}\|_2 \leq C\sigma^{-1/2} \|\ell_{c,-}\|_1$$

by the finite eigenrelation and (8). Hence

$$\|CP_{c,-}\|_{\mathfrak{S}_2} = \|Cr_{c,-}\|_2 \|\ell_{c,-}\|_2 \leq Ch\sigma^{-3/2}.$$

Divide by the C lower bound. □

Corollary 5.2 (RH-50 residue premise). *The explicit Perron/parity residue part of the RH-50 dyadic Hardy-control condition is uniformly bounded. Neither $\|\ell_d\| = O(h\sigma^{-1})$ nor a polylogarithmic upper for the complete finite projector is required for this range-restricted result.*

Remark 5.3. Multiplying the separate weak-factor smoothing bounds gives only $\|P_{n,-}\|_{L^2 \rightarrow L^2} = O(\sigma^{-1/2})$. The theorem avoids this coarse global estimate: each range action gains a complementary power before the factors are multiplied.

6 A sharp half-power barrier

Theorem 6.1 (Sharp weak-information detail scale). *There are $c, C, a_0, \sigma_0 > 0$ such that*

$$ch\sigma^{-3/2} \leq \|Q_n \mathcal{P}_\sigma\|_{L^1 \rightarrow L^2} \leq Ch\sigma^{-3/2} \tag{13}$$

when $0 < \sigma < \sigma_0$ and $h/\sigma < a_0$. Therefore uniform left L^1 information alone cannot yield $\|\ell_d\| = O(h\sigma^{-1})$.

Proof. The upper bound is (6). Choose an interior source point x_0 and approximate a unit point mass there by L^1 densities. Their images converge in L^2 to the Gaussian row $k_\sigma(x_0, \cdot)$. On an interior target window,

$$\|\partial_y k_\sigma(x_0, \cdot)\|_2 \geq c\sigma^{-3/2}.$$

The adjacent Haar Taylor expansion, summed over cells in that window, gives

$$\|Q_n k_\sigma(x_0, \cdot)\|_2 \geq ch\sigma^{-3/2}$$

when h/σ is small. Taking the operator supremum proves the lower bound. □

The physical factors are not arbitrary L^1 inputs. Their postcritical weights produce structure that this operator norm cannot see. RH-50 proved the continuum rounded-spike derivative scale $\Theta(\sigma^{-1})$. Transferring it to intrinsic finite factors may recover the optional $O(\sqrt{\sigma})$ residue law, but theorem 6.1 identifies the extra information required.

7 Five-scale intrinsic-factor audit

The experiment uses the sparse eight-sigma-cutoff family and exact Haar nesting of RH-49 and RH-50. The fine resolution is

$$N\sigma = 20.48,$$

and the largest dimension is 40960. This fixed resolution is a production diagnostic, not a realization of $n\sigma^2 \rightarrow \infty$.

σ	N	weak cond.	Perron detail	parity detail	fine +	fine -	right -
0.0100	2048	1.0968	0.0441	0.0553	0.0462	0.0574	0.0315
0.0040	5120	1.0765	0.0422	0.0511	0.0281	0.0339	0.0130
0.0020	10240	1.0604	0.0411	0.0489	0.0195	0.0230	0.00675
0.0010	20480	1.0485	0.0405	0.0475	0.0136	0.0159	0.00356
0.0005	40960	1.0385	0.0401	0.0465	0.00951	0.0110	0.00191

Table 1: Five-scale audit. The weak condition is $\|r\|_\infty \|\ell\|_1$ for parity. Detail columns are normalized by $h\sigma^{-1}$; the final columns are residue actions.

The parity weak products are

$$1.0968, 1.0765, 1.0604, 1.0485, 1.0385,$$

while the Perron products equal one to rounding. The parity values of $\|P\|_-^2 / \log(1/\sigma)$ decrease from 0.272 to 0.208.

The finite detail is much sharper than the weak theorem:

$$0.0401 \leq \frac{\|\ell_{f,+},d\|_2}{h\sigma^{-1}} \leq 0.0441, \quad 0.0465 \leq \frac{\|\ell_{f,-},d\|_2}{h\sigma^{-1}} \leq 0.0553.$$

The fitted powers of these ratios are slightly positive, so no growth is observed.

The direct residue fits are

$$\text{fine Perron : } \sigma^{0.52752}, \quad \text{fine parity : } \sigma^{0.55003}, \quad (14)$$

$$\text{right parity : } \sigma^{0.93624}. \quad (15)$$

The right Perron action is below 2.6×10^{-15} at every scale. The normalized coupling constants are nearly fixed:

$$0.0941 \leq \frac{\|B\|_{\mathfrak{S}_2}}{h\sigma^{-3/2}} \leq 0.0955, \quad 0.16714 \leq \frac{\|C\|_{\mathfrak{S}_2}}{h\sigma^{-3/2}} \leq 0.16715.$$

One-level parity comparisons have maximum left L^1 error 8.21×10^{-5} , right L^∞ error 2.90×10^{-4} , and relative projector defect 1.46×10^{-4} .

Intrinsic peripheral residues: weak-factor closure and stronger observed transfer

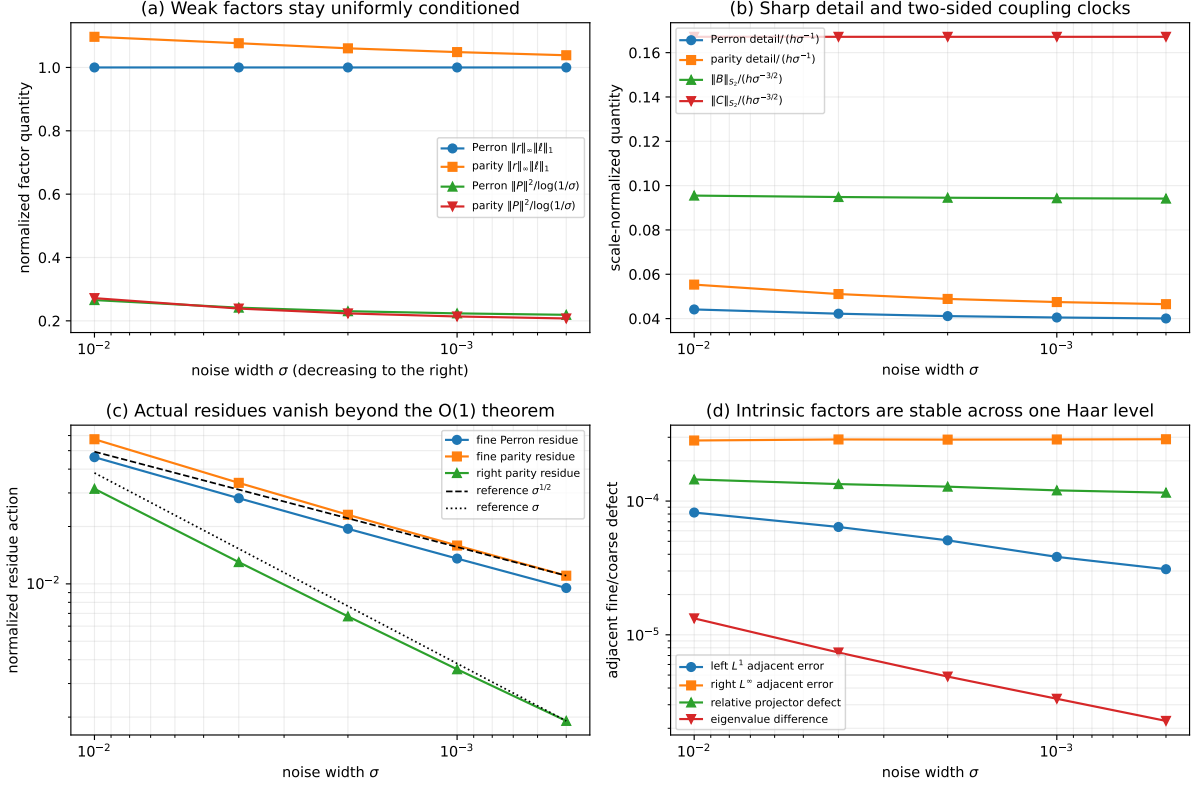


Figure 1: Intrinsic peripheral transfer audit. (a) Weak factors remain conditioned. (b) Actual details follow $h\sigma^{-1}$, while both couplings follow $h\sigma^{-3/2}$. (c) All residues vanish, stronger than the analytic $O(1)$ theorem. (d) Adjacent intrinsic factors remain stable across one Haar level.

8 Roadmap consequence

The original A2 milestone requested sharp finite-detail transfer and a polylogarithmic coarse parity projector. [Theorem 5.1](#) replaces both with weaker factor information and a stronger targeted conclusion: every residue action entering RH-50 is $O(1)$.

Sufficient A2 residue gate

Closed analytically on the adopted $n\sigma^2 \rightarrow \infty$ schedule.

Original $O(\sqrt{\sigma})$ target

Not closed analytically. It is supported by the audit but blocked from weak-norm-only proofs by [theorem 6.1](#).

Complete finite projector

No longer required for the range-restricted RH-50 residue ledger.

Remaining Stage A work

A1 still needs a growing-horizon Hardy/Stein trace budget. A3 still needs deterministic or interval infinite-tail and sparse-cutoff validation.

The next scheduled paper is RH-53, focused on the infinite Hardy tail and the sparse/full Gaussian cutoff bridge.

8.1 No arithmetic or Hilbert–Pólya conclusion

The TPC series remains an independent twin-prime and prime-correlation program and is not an assumption here. Nothing in this paper constructs a prime-power trace formula, identifies a zeta zero, builds a self-adjoint Hilbert–Pólya operator, derives a $T \log T$ law, proves the Riemann hypothesis, or proves a twin-prime statement.

9 Reproducibility and theorem boundary

The archive contains the algebra in `src/factor_transfer`, the pilot and certificate builders in `experiments`, the figure, tests, result ledgers, dependency hashes, and archive verification. Principal commands:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q
OPENBLAS_NUM_THREADS=16 OMP_NUM_THREADS=16 \
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/run_factor_transfer_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/build_factor_transfer_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/verify_archive.py
```

The weak-factor theorem, direct kernel bounds, two-sided coupling scales, uniform residue closure, and half-power barrier are analytic. The sharper intrinsic $h\sigma^{-1}$ detail law, vanishing residue powers, and adjacent defects are floating binary64 evidence. No interval constant is claimed for the stored matrices.

References

- [1] Abraham Boyarsky and Paweł Góra. *Laws of Chaos: Invariant Measures and Dynamical Systems in One Dimension*. Birkhäuser, Boston, 1997. doi: 10.1007/978-1-4612-2024-9.
- [2] Gerhard Keller and Carlangelo Liverani. Stability of the spectrum for transfer operators. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 28(1):141–152, 1999. URL http://www.numdam.org/item/ASNSP_1999_4_28_1_141_0/.
- [3] Tien-Yien Li. Finite approximation for the Frobenius–Perron operator: A solution to Ulam’s conjecture. *Journal of Approximation Theory*, 17(2):177–186, 1976. doi: 10.1016/0021-9045(76)90037-X.
- [4] Liang Wang. Square-root parity splitting at a quadratic band-merging map: Coupled critical-value boundary layers and a corrected small-noise law. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-14-square-root-parity-boundary-layer.
- [5] Liang Wang. Residue-deflated directional resolvents at a quadratic critical fold: A quarter-power stable-rank bridge for intrinsic Riesz identification. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-49-directional-reduced-resolvent.

- [6] Liang Wang. Two-pole Hilbert–Schmidt hardy energies for small-noise range resolvents: Directional Stein certificates and a global-contraction no-go. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-50-two-pole-hilbert-schmidt-hardy.
- [7] Liang Wang. Logarithmic peripheral conditioning and an anchored small-noise bulk mesh law. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-47-logarithmic-peripheral-conditioning.
- [8] Liang Wang. Cyclic-rank obstructions and growing-horizon Stein certificates: Minimal gramians, conic no-go witnesses, and a viable block route. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-51-cyclic-rank-growing-horizon-stein.